

Rolls-Royce Trent 800

Developed from the earlier RB211 powerplant, the Trent 800 turbofan was designed specifically for the Boeing 777. Work on the high-bypass-ratio engine began in 1993 and the first Trent-powered 777 entered service with Thai Airways in March 1996. Today more than 40 percent of all 777s in service are powered by Trent 800 engines.

POWERING THE 777

When Boeing announced a larger version of its 767 model in the 1980s, Rolls-Royce proposed that the Trent 760 engine could be used. When the 767X project was abandoned in favor of the 777, the need for an even larger engine became apparent, and the Trent 800 was the result. The turbofan runs a fan, an intermediate compressor, and an HP compressor on three separate shafts, allowing the blade velocities to be optimized. In January 1994, the Trent 800 demonstrated a world-record 106,087 lb (471 kN) thrust.

ENGINE SPECIFICATIONS	
Dates produced	1993 to present
Configuration	High-bypass-ratio turbofan
Fuel	Jet fuel
Power output	93,400 lb (415 kN) thrust
Weight	13,825 lb (6,270 kg)
Compressors	8-stage IP, 6-stage HP
Turbines	Single-stage HP, single-stage IP, 5-stage LP
Combustor	Single annular combustor with 24 fuel injectors



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Tail cone
This smooths airflow from the rear of the turbine assembly.

Low pressure
LP turbine module

Exhaust casing

Three shafts
The three-shaft layout of the Trent 800 adds mechanical complexity but enables the engine to develop high thrust from a shorter, lighter engine than earlier two-shaft turbofans.